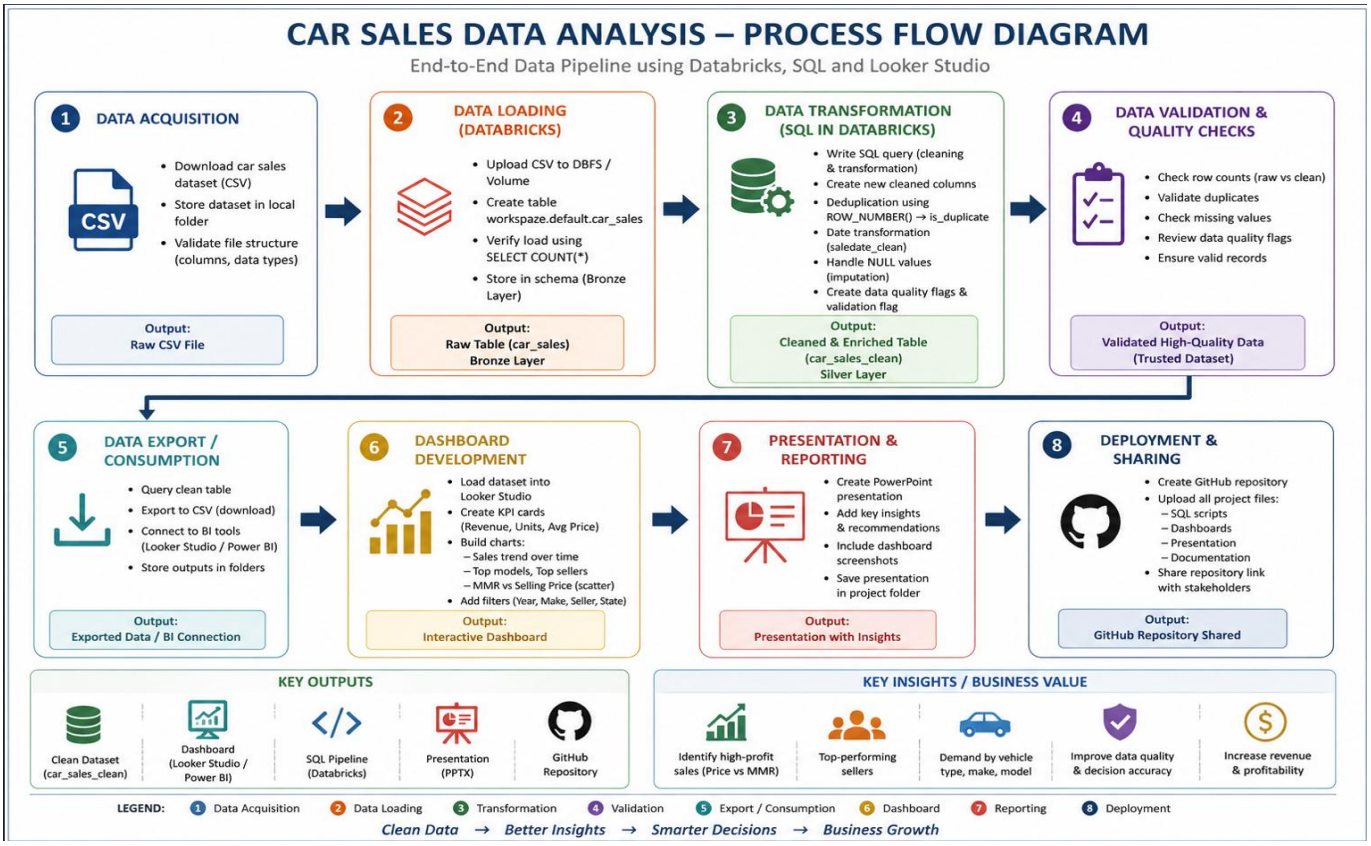
CAR SALES DATA ANALYSIS – PROCESS FLOW DIAGRAM
End-to-End Data Pipeline using Databricks, SQL and Looker Studio
1 DATA ACQUISITION
CSV
• Download car sales dataset (CSV)
• Store dataset in local folder
• Validate file structure (columns, data types)
Output: Raw CSV File
2 DATA LOADING (DATABRICKS)
• Upload CSV to DBFS / Volume
• Create table workspaze.default.car_sales
• Verify load using SELECT COUNT(*)
• Store in schema (Bronze Layer)
Output: Raw Table (car_sales) Bronze Layer
3 DATA TRANSFORMATION (SQL IN DATABRICKS)
• Write SQL query (cleaning & transformation)
• Create new cleaned columns
• Deduplication using ROW_NUMBER() → is_duplicate
• Date transformation (saledate_clean)
• Handle NULL values (imputation)
• Create data quality flags & validation flag
Output: Cleaned & Enriched Table (car_sales_clean) Silver Layer
4 DATA VALIDATION & QUALITY CHECKS
• Check row counts (raw vs clean)
• Validate duplicates
• Check missing values
• Review data quality flags
• Ensure valid records
Output: Validated High-Quality Data (Trusted Dataset)
5 DATA EXPORT / CONSUMPTION
• Query clean table
• Export to CSV (download)
• Connect to BI tools (Looker Studio / Power BI)
• Store outputs in folders
Output: Exported Data / BI Connection
6 DASHBOARD DEVELOPMENT
• Load dataset into Looker Studio
• Create KPI cards (Revenue, Units, Avg Price)
• Build charts:
– Sales trend over time
– Top models, Top sellers
– MMR vs Selling Price (scatter)
• Add filters (Year, Make, Seller, State)
Output: Interactive Dashboard
7 PRESENTATION & REPORTING
• Create PowerPoint presentation
• Add key insights & recommendations
• Include dashboard screenshots
• Save presentation in project folder
Output: Presentation with Insights
8 DEPLOYMENT & SHARING
• Create GitHub repository
• Upload all project files:
– SQL scripts
– Dashboards
– Presentation
– Documentation
• Share repository link with stakeholders
Output: GitHub Repository Shared
KEY OUTPUTS
Clean Dataset (car_sales_clean)
Dashboard (Looker Studio / Power BI)
SQL Pipeline (Databricks)
Presentation (PPTX)
GitHub Repository
KEY INSIGHTS / BUSINESS VALUE
Identify high-profit sales (Price vs MMR)
Top-performing sellers
Demand by vehicle type, make, model
Improve data quality & decision accuracy
Increase revenue & profitability
LEGEND: 1 Data Acquisition 2 Data Loading 3 Transformation 4 Validation 5 Export / Consumption 6 Dashboard 7 Reporting 8 Deployment
Clean Data → Better Insights → Smarter Decisions → Business Growth